

# Data Quality Report

**Project:** US Accidents Dataset – Real-World Data Engineering Challenge

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**Course:** Data Engineering

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## Objective

The objective of this project is to transform the raw US Accidents dataset into a clean, structured, and analysis-ready dataset using data engineering techniques. The project includes data cleaning, feature engineering, categorical encoding, numerical feature scaling, validation, and the development of a reusable preprocessing pipeline.

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## Dataset Information

| Item                 | Value                       |
|----------------------|-----------------------------|
| Dataset Name         | US Accidents Dataset        |
| Source               | Kaggle                      |
| Original Format      | CSV                         |
| Final Format         | CSV                         |
| Programming Language | Python                      |
| IDE                  | Visual Studio Code          |
| Libraries Used       | Pandas, NumPy, Scikit-learn |

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# Data Audit Summary

## Task 1 – Data Audit

The dataset was examined before preprocessing to identify quality issues.

The following checks were performed:

- Missing values
  - Duplicate records
  - Data types
  - Memory usage
  - Invalid coordinates
  - Constant columns
  - Low variance columns
  - High-cardinality categorical columns
  - Rare categories
  - Date and time consistency
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# Data Cleaning Summary

## Task 2 – Data Cleaning

The following preprocessing steps were performed:

- Missing numerical values filled using the median.
- Missing categorical values filled using the mode.
- Date columns converted to datetime format.
- Forward filling applied to missing datetime values.
- Duplicate records removed.
- Invalid latitude and longitude values removed.
- Text values standardized using title case.
- Outliers detected using the IQR method.
- Impossible values such as negative distances removed.
- Records with invalid accident duration removed.

# Feature Engineering

## Task 3 – Feature Engineering

The project includes the creation of **30 new features** to improve the quality and usefulness of the dataset for future analysis.

The engineered features are grouped into different categories:

### Time Features

- Hour
- Day
- Month
- Quarter
- Weekday
- Weekend Flag
- Rush Hour Indicator
- Night Indicator
- Holiday Indicator

### Weather Features

- Temperature Category
- Visibility Category
- Wind Category
- Rain Indicator
- Severe Weather Indicator

### Geographic Features

- State Frequency
- City Frequency
- Distance Category
- Urban/Rural Flag
- State Code Length

### Traffic Features

- Accident Duration (Minutes)
- Traffic Severity Score
- Road Complexity Score
- Traffic Control Score
- Road Hazard Score

### Text Features

- Description Length
- Word Count
- Average Word Length
- Accident Keyword Indicator
- Weather Keyword Indicator
- Road Closure Keyword Indicator

**Total Engineered Features: 30**

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## Task 4 – Encoding

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Different encoding techniques were applied to convert categorical variables into numerical form.

#### Label Encoding

Applied to:

- Weather\_Condition

#### One-Hot Encoding

Applied to:

- Source

#### Frequency Encoding

Applied to:

- State

#### Target Encoding

Target Encoding was not applied because the assignment focuses only on preprocessing and does not require a target variable.

#### Hash Encoding

Hash Encoding was studied as an optional encoding technique but was not implemented because the selected encoding methods were sufficient for this dataset.

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## Task 5 – Scaling

### Task 5 – Feature Scaling

Several scaling techniques were explored and compared during the project.

The following methods were implemented:

- Min-Max Scaling
- Standard Scaling
- Robust Scaling
- Max Absolute Scaling

For the final preprocessing pipeline, **Max Absolute Scaling** was selected because it is memory-efficient and suitable for very large datasets while preserving sparse data characteristics.

The following numerical columns were scaled:

- Temperature(F)
  - Visibility(mi)
  - Wind\_Speed(mph)
  - Distance(mi)
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## Task 6 – Pipeline Validation

### Task 6 – Pipeline Validation

The preprocessing pipeline automatically performs the following steps:

- Load raw dataset
- Clean the data
- Generate engineered features
- Encode categorical variables

- Scale numerical features
- Validate the processed dataset
- Export the final production-ready dataset

Validation checks include:

- Total number of rows
  - Total number of columns
  - Missing value summary
  - Duplicate record summary
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# Final Dataset Statistics

Use the values from your pipeline output:

| Metric                 | Value                |
|------------------------|----------------------|
| Rows                   | 6,764,788            |
| Columns                | 81                   |
| Duplicate Rows         | 0                    |
| Engineered Features    | 30                   |
| Encoding Techniques    | 3                    |
| Scaling Technique Used | Max Absolute Scaling |

For **Missing Values**, use the value shown in your pipeline output:

**Missing Values:** 4,125,557

**Note:** These remaining missing values are expected because some attributes (for example, weather-related fields or airport information) are genuinely unavailable in the original dataset. They were not all removed to preserve as much useful information as possible.

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## Conclusion

This project successfully transformed the raw US Accidents dataset into a production-ready dataset through a complete preprocessing pipeline. The pipeline performs automated data cleaning, feature engineering, categorical encoding, numerical scaling, validation, and dataset export. The modular design makes the solution reusable, maintainable, and suitable for future data engineering and data science workflows.